

SOHAM HARIDAS GAIKWAD

B.Tech CSE • SPIT Mumbai • 3rd Year (2024–2028)
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[GitHub](#) • [Portfolio](#) • [LeetCode](#)

EDUCATION

B.Tech – Computer Science & Engineering

2024 – 2028

Sardar Patel Institute of Technology (SPIT), Mumbai

CGPA: 8.09 | Coursework: Data Structures & Algorithms, Operating Systems, Computer Networks, DBMS, Computer Architecture, Data Science

TECHNICAL SKILLS

Programming Languages: Python, C, Java

Machine Learning / Deep Learning / LLM: Machine Learning, Deep Learning (Neural Networks), Large Language Models (LLM), Retrieval-Augmented Generation (RAG), LangChain, Google Gemini API, FAISS (Vector Search), Prompt Engineering, Conversational Memory Design, Supervised Learning, Gradient Boosting, XGBoost, LightGBM, Hyperparameter Tuning, Predictive Analytics, Time Series Forecasting (Prophet), PyTorch, TensorFlow, Keras, Scikit-learn

Data Science & Analysis: Exploratory Data Analysis (EDA), Feature Engineering, Data Cleaning, Data Processing, Pandas, NumPy, Matplotlib, Seaborn, Jupyter Notebooks

NLP & AI Applications: Sentiment Analysis (VADER), Named Entity Recognition, Multilingual NLP (langdetect), Multimodal AI (Text + Image Reasoning), Semantic Search, Retrieval-Grounded Chatbots

Web Development & Deployment: HTML, CSS, Flask, Streamlit, Web Scraping

Databases: MySQL, Database Management Systems (DBMS)

CS Core: Data Structures & Algorithms, Operating Systems, Computer Networks, Computer Architecture, Object-Oriented Programming (OOP)

Tools & DevOps: Git, GitHub, GitHub Actions, CI/CD (basics), VS Code, Linux, Windows, Conda/Environment Management

UI / UX: Figma (Basic), Wireframing, User-Centered Design

Interpersonal: Problem Solving

PROJECTS

Real-Time Gen AI Customer Service Bot — 6-Feature LLM System

ElevanceSkills Internship | 2026

- Built and extended a retrieval-augmented customer service chatbot (Google Gemini + LangChain + FAISS) into 6 production-style AI systems on top of a shared architecture.
- **Sentiment-Aware Chatbot:** integrated VADER sentiment classification to detect customer emotion and dynamically adapt response tone in real time.
- **Medical Q&A Chatbot:** built a custom XML parser to process 16,000+ MedQuAD Q&A pairs; implemented FAISS-based retrieval and keyword-driven medical entity recognition (symptoms/treatments).
- **Dynamic Knowledge Base:** designed an incremental vector-index update mechanism to add new information without full re-embedding.
- **arXiv Research Assistant:** built a paper-search and summarization system over live arXiv API data, with conversational memory and query-rewriting to correctly resolve follow-up questions.
- **Multimodal Assistant:** implemented text + image reasoning using Gemini's native multimodal API with persistent conversational memory across turns.
- **Multilingual Chatbot:** built automatic language detection and cross-lingual response generation (verified across English, Hindi, Marathi, Spanish) with context retention across language switches.

Drive-Shield — Real-Time Edge ADAS & Incident Forensics Platform

Personal Project | 2026

Tech Stack: Python, PyTorch, YOLOv8, OpenCV, FastAPI, WebSockets, SORT, SQLite, JavaScript

- Developed an edge-computing ADAS pipeline using YOLOv8 and SORT tracking, achieving 30 FPS inference (<45ms latency) for multi-class obstacle detection, monocular depth estimation, and optical flow speed tracking.

- Engineered predictive risk algorithms evaluating Time-to-Collision ($TTC \leq 2.0s$), abrupt deceleration ($>6.5 \text{ m/s}^2$), and pedestrian fall/tumble postures to trigger millisecond safety warnings.
- Built a FastAPI & WebSocket dual-stream cockpit UI with an in-memory rolling buffer, automatically capturing and archiving 30-second pre/post-incident forensic video evidence signed with SHA-256 cryptographic hashes.

BIT-SHIELD — Bitcoin Forensics & Anomaly Detection Engine

Personal Project | 2026

Tech Stack: Python, FastAPI, NetworkX, Scikit-Learn, React 18, TypeScript, Cytoscape.js, SQLite

- Engineered an air-gapped, zero-cloud Bitcoin transaction forensics platform featuring a universal offline data fabric with fuzzy regex schema normalization for arbitrary CSV/JSON datasets and automated SHA-256 chain-of-custody verification.
- Implemented an algorithmic graph mining engine detecting 8 behavioral money-laundering motifs (Rapid-Hop, CoinJoin mixing, Peel Chains, High Fan-Out/In) across directed bipartite transaction topologies (Address $\leftrightarrow$ Transaction).
- Built a 5-lens risk fusion pipeline fusing deterministic rules (R1-R8) with unsupervised Isolation Forest scoring, generating explainable, court-admissible evidence chains.
- Designed an interactive SOC forensic workstation with Cytoscape.js supporting 4 dynamic graph layouts, live entity search, and 2x high-resolution PNG evidence export.

RAG-Based AI Teaching Assistant

Personal Project | 2025

- Built a Retrieval-Augmented Generation pipeline that answers course questions grounded in uploaded study material.
- Combined an LLM with a vector-based retrieval layer to reduce hallucination and cite source content.

HyperLocal Demand Forecaster V2

Data Science Course Project | 2025

- Ensemble model (LightGBM + XGBoost + CatBoost) with Optuna hyperparameter tuning and Ridge meta-learner stacking.
- Built a SHAP explainability layer and Indian-festival-proximity features to capture local demand patterns.

FIFA World Cup Linear Regression Analysis

Data Science Course | 2025

- Performed multicollinearity/VIF analysis with a self-contained JupyterLab notebook, including clean EDA and model interpretation.

CERTIFICATIONS & COURSES

Real-Time Gen AI Project Internship — Customer Service Bot

ElevanceSkills | Jul 2026

- Mentor-guided internship extending a base LLM chatbot into 6 production-style AI systems (sentiment analysis, medical Q&A, dynamic knowledge base, research assistant, multimodal AI, multilingual support).
- Training Certificate: [View Certificate](#)
- Internship Completion Certificate: [View Certificate](#)

Data Science with Python

CodeWithHarry | Mar 2026

- Covered end-to-end data science workflows — EDA, ML model building with Scikit-learn, and reproducible research.
- Hands-on with Pandas, NumPy, Matplotlib; explored neural network fundamentals and LLM concepts.
- Certificate: [View Certificate](#)

ACADEMIC HIGHLIGHTS

- CGPA 8.09 in a competitive CSE program at SPIT — consistently pushing for higher scores.
- Strong in core subjects: OS, Computer Networks (subnetting, TCP/IP, routing), DAA, DBMS.
- Self-learning Git, DevOps basics, and GitHub Actions alongside coursework.
- Built and deployed 6 end-to-end LLM-based systems during a real-time Gen AI internship, in addition to independent data science and RAG projects.